Monologue (Topic: Weber-Fechner law)

Speaker:

大家好，今天我们要探讨的是心理学中的一个重要定律——韦伯-费希纳定律。你有没有想过，为什么我们对刺激的感知不是简单地随着刺激强度的增加而线性变化？这正是韦伯和费希纳定律要揭示的秘密。

Hello everyone, today we are going to explore an important law in psychology—the Weber-Fechner law. Have you ever wondered why our perception of a stimulus does not increase linearly with the intensity of the stimulus? This is exactly the insight that the Weber-Fechner law reveals.

首先，韦伯告诉我们，感知的变化并不依赖于刺激的绝对差异，而是依赖于相对差异。换句话说，最小可觉察差异和原始刺激强度的比值是一个常数。大家可以想象一下，如果你手里拿着 100 克的重物，增加 2 克你可能能感觉到变化，但如果是拿 200 克的重物，你需要增加 4 克才能感觉到差异。

First, Weber tells us that changes in perception do not depend on the absolute difference of the stimulus, but on the relative difference. In other words, the ratio of the just noticeable difference to the original stimulus intensity is constant. Imagine, for instance, that if you hold a 100-gram weight, adding 2 grams might be noticeable, but if you hold a 200-gram weight, you would need to add 4 grams to perceive the difference.

这个现象是韦伯通过经典的重物辨别实验得出的。实验中，被试者先后提起两个重物，判断哪个更重。结果显示，无论重物的绝对重量是多少，最小可觉察的差异和原始重量的比例基本保持不变。

This phenomenon was discovered by Weber through a classic weight discrimination experiment. Participants lifted two weights sequentially and judged which one was heavier. The results showed that regardless of the absolute weight, the ratio of the just noticeable difference to the original weight remained approximately constant.

嗯，这是不是很有趣？这说明我们感知的敏感度是相对的，而不是绝对的。换句话说，我们的大脑是在比较差异的比例，而不是单纯的数值。

Isn’t that interesting? This shows that our perceptual sensitivity is relative rather than absolute. In other words, our brain compares the proportion of difference rather than just the numerical value.

接下来，费希纳则是在韦伯定律基础上的进一步发展。费希纳通过数学推导，发现心理感觉强度和物理刺激强度之间是对数关系。也就是说，当物理刺激强度增加时，心理感觉的增加速度会逐渐减缓。

Next, Fechner built upon Weber’s law and developed it further. Through mathematical derivation, Fechner found that the perceived intensity is logarithmically related to the physical stimulus intensity. That is, as the physical stimulus increases, the rate of increase in perceived sensation gradually slows.

举个例子，在嘈杂的环境中，你需要把音量大幅提高，才能明显感觉到声音变大。因为声音的心理感受并不是线性增加的，而是呈现出一种‘递减的边际效应’。

For example, in a noisy environment, you need to increase the volume significantly in order to perceive a noticeable increase in loudness. This is because the psychological perception of sound does not increase linearly but exhibits a “diminishing marginal effect.”

你们有没有注意过手机音量调节？如果音量是线性增加的，到了高音量阶段，细微的调整几乎察觉不到。其实设计师们正是利用了费希纳定律，采用对数尺度来设计音量控制，让用户在不同音量下都能感受到明显的变化。

Have you noticed mobile phone volume controls? If the volume increased linearly, subtle adjustments at high levels would hardly be noticeable. Designers actually utilize the Fechner law, applying a logarithmic scale for volume control so that users can perceive noticeable changes at all volume levels.

不仅是声音，视觉设计中也可以应用这些定律。比如屏幕亮度的调节，如果调整幅度太小，用户根本察觉不到变化；太大又会让人觉得不舒服。韦伯定律中的常数 K 值就能帮我们确定最小可觉察的差异，指导设计更符合用户感知。

These laws are not limited to sound; they can also be applied in visual design. For example, when adjusting screen brightness, if the change is too small, users cannot notice it; if too large, it may feel uncomfortable. The constant K in Weber’s law helps determine the just noticeable difference, guiding designs that better match user perception.

嗯，大家可以想象一下，如果设计师忽视了这些感知规律，产品的反馈可能会让用户感到迟钝或过于敏感，影响使用体验。

You can imagine that if designers ignore these perceptual laws, product feedback may feel either dull or overly sensitive to users, affecting the user experience.

那么，韦伯-费希纳定律给我们的最大启示是什么呢？其实是告诉我们，设计的有效性不在于物理量的精确控制，而在于满足用户的感知规律。

So, what is the key takeaway from the Weber-Fechner law? It tells us that the effectiveness of design does not lie in precise physical control, but in aligning with users’ perceptual principles.

换句话说，我们要关注的是用户能否察觉到变化，而不是变化本身有多大。比如，一个按钮的颜色变化，如果变化太小，用户可能根本看不到；变化太大，又可能让界面显得突兀。

In other words, we should focus on whether users can perceive the change, rather than how large the change itself is. For example, if a button’s color change is too subtle, users might not notice it; if it is too drastic, it could make the interface appear abrupt.

这让我想起一个日常例子：你有没有发现，早晨的阳光明明比晚上强很多，但你感觉的亮度差异却没那么大？这也是因为心理感知和物理刺激之间的非线性关系。

This reminds me of a daily example: have you noticed that morning sunlight is much stronger than evening sunlight, yet the perceived difference in brightness doesn’t feel that large? This is due to the non-linear relationship between psychological perception and physical stimulus.

嗯，回到我们的主题，韦伯-费希纳定律揭示了感知的相对性和对数关系，这对于理解人类感官和设计界面都非常重要。

Returning to our topic, the Weber-Fechner law reveals the relativity and logarithmic relationship of perception, which is very important for understanding human senses and interface design.

其实，这些定律不仅仅是理论，它们在我们的生活中无处不在。比如调节音量、调整亮度、甚至在广告设计中，如何让颜色和大小的变化更显著，都是基于这些感知规律。

In fact, these laws are not just theoretical; they are everywhere in our daily lives. For instance, adjusting volume or screen brightness, or even making color and size changes more noticeable in advertising design, all rely on these perceptual principles.

那么，大家有没有想过，未来智能设备还能如何利用这些定律来提升用户体验呢？比如，个性化调节反馈强度，满足不同用户的感知差异。

So, have you ever thought about how future smart devices could use these laws to enhance user experience? For example, by personalizing feedback intensity to match different users’ perceptual sensitivity.

嗯，想象一下，如果设备能根据你的感知敏感度自动调整反馈强度，是不是更贴心、更智能呢？这也是心理物理学应用的一个潜在方向。

Yes, imagine if a device could automatically adjust feedback intensity based on your perceptual sensitivity. Wouldn’t that be more considerate and intelligent? This is also a potential application of psychophysics.

总结一下，韦伯告诉我们感知差异的比例恒定，而费希纳揭示了心理感觉与物理刺激的对数关系。两者共同解释了我们对刺激变化的感知机制。

In summary, Weber tells us that the ratio of perceptual difference is constant, while Fechner reveals the logarithmic relationship between psychological sensation and physical stimulus. Together, they explain how we perceive changes in stimuli.

该定律不仅丰富了我们对人类感官的理解，也为设计师提供了科学依据，帮助我们设计出更符合用户感知的产品。

These laws not only enrich our understanding of human perception, but also provide designers with scientific guidance to create products that better align with user perception.

好了，今天的分享就到这里。希望大家能把韦伯-费希纳定律记在心里，下次设计或者体验产品时，能更好地理解背后的感知秘密。

That’s all for today’s discussion. I hope you keep the Weber-Fechner law in mind so that next time you design or experience a product, you can better understand the perceptual principles behind it.

谢谢大家的聆听，有没有什么问题或者想法，欢迎随时提出来，我们一起讨论。

Thank you all for listening. If you have any questions or thoughts, feel free to share them, and we can discuss together.